

Infrared (IR) sensors detect infrared radiation emitted by objects. These sensors are versatile and are used in a variety of applications including motion detection, temperature measurement, and remote-control systems.

Key Features

- Detects infrared radiation for motion, temperature, and remote-control applications.
- Wide detection range adaptable from centimeters to several meters.
- Multiple output modes (analog, digital, PWM) for flexible system integration.
- Fast response time suitable for both real-time control and monitoring.
- Wide field of view for broader coverage or narrow FOV for precision sensing.
- Operates across industrial temperature ranges for robust applications.
- Compact package options for easy integration in various devices.



IR Sensor Board

Technical Specification

Parameter	Value
Operating Voltage	3.3 V to 5 V (higher for some variants)
Current Consumption	Few mA to tens of mA
Detection Range	Few cm to several meters (depends on sensor type)
Field of View (FOV)	30° – 180°
Wavelength Range	Near-IR: 700 nm – 1400 nm Thermal IR: 8 μm – 14 μm
Response Time	Microseconds to milliseconds
Output Types	Analog, Digital, PWM
Operating Temperature	−40 °C to +85 °C
Package Types	Through-hole, SMD, TO-5, TO-18, etc.

Applications

- **Motion Detection** — Security systems, automatic lighting, robotics.
- **Temperature Measurement** — Contactless thermometers, industrial monitoring.
- **Remote Controls** — TVs, AC units, and consumer electronics.
- **Gesture Recognition** — Smart devices, interactive displays.
- **Proximity Sensing** — Smartphones, touchless switches.
- **Flame Detection** — Fire alarms and safety systems.
- **Medical Devices** — Thermal imaging, patient monitoring.
- **Industrial Automation** — Object counting, positioning systems.

- **Environmental Monitoring** — Gas/flame sensing with thermal IR.
- **Automotive** — Night vision systems, driver assistance sensors.